OncoEvidence: Mechanism-Guided Evidence Triage for Cancer Drug Repurposing

When a knowledge graph earns its place, and how to show its mechanism reasoning is predictive and faithful

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Abstract. OncoEvidence ranks oncology drug-repurposing candidates on the PrimeKG knowledge graph, extracts the multi-hop mechanism path that would explain each one, and uses a literature-grounded language model to grade whether that mechanism is supported. The project is built around an honest question: once a strong tabular ranker is in place, what does the graph actually add? We show the answer is *mechanism*, and we then stress that mechanism signal along the two axes that matter for a real triage tool. On a corrected, leakage-safe benchmark (5 seeds) a tuned XGBoost matches or beats the graph network at link ranking. But only the graph yields a traceable drug → target → cancer-gene → cancer path; that path structure separates true indications from random pairs at AUROC 0.879 and agrees with curated DrugMechDB at 0.802. Two new results make the case stronger. On a prospective time-split the graph model ranks *future* oncology indications at AUROC 0.930 and beats a structure-blind control by +0.047; concretely, trained only on pre-2005 structure it surfaces 11× as many future indications in the top 50 as the control, including post-cutoff approvals like romidepsin and pralatrexate for cutaneous T-cell lymphoma and nilotinib for CML. And a counterfactual test shows the mechanism head is *faithful*: deleting the curated mechanism edge, and only that edge, collapses the prediction ($p \approx 3.6 \times 10^{-34}$). An orthogonal check against ClinicalTrials.gov is reported as a fully characterized negative: after removing both drug and cancer popularity, the interaction signal vanishes (AUROC 0.475).

Six findings at a glance.

1. Plain link ranking does not need the graph: tuned XGBoost matches or beats the GNN in every regime.
2. Only the graph produces a traceable multi-hop mechanism path.
3. Mechanism structure separates true indications from random pairs (AUROC 0.879), survives hard negatives, and matches DrugMechDB (0.802).
4. Blinded mechanism recovery: with the direct edge hidden, only the joint GNN names the bridge gene (R@10 0.25 vs ∼0 baselines).
5. **New, prospective:** on a time-split the GNN ranks future indications at AUROC 0.930 and surfaces 11× as many in the top 50 as a structure-blind control (named post-2005 approvals included).
6. **New, faithfulness:** deleting the curated mechanism edge hurts ≫ a random edge (contrast 0.124, $p \approx 3.6 \times 10^{-34}$, faithful 83%).

1 Background and significance

Developing a new drug takes many years and large cost, while thousands of approved drugs already carry known safety profiles. If an approved drug acts on a protein or pathway that also drives a cancer, repurposing it is a fast route to a candidate therapy. Biomedical knowledge graphs such as PrimeKG encode millions of relationships among drugs, proteins, genes, pathways, and diseases, so the information needed to reason about such mechanisms already exists as paths through the graph. The open problem is *trust*: a bare drug-disease score hides its reasoning and cannot separate a real mechanism from a coincidence.

Novelty, stated carefully. Knowledge-graph repurposing with explanations is not new: TxGNN [1] is a graph foundation model with multi-hop rationales, and KGML-xDTD [3] predicts treatment links with KG-based mechanism paths. This project does not claim "a knowledge graph plus a language model for repurposing" as novel. The defensible contribution is methodological: quantify *when* graph structure adds value beyond a strong tabular ranker, show that its value is mechanism extraction rather than better ranking, and audit those mechanisms three ways most demos skip — against a curated mechanism database, against a prospective time-split, and against a counterfactual faithfulness test.

2 Data and system

Knowledge graph. We use PrimeKG [2], processed into a heterogeneous graph of 129,312 nodes and 8,100,498 directed edges over 50 relation types (Table 1). The prediction target is the drug $\xrightarrow{\text{indication}}$ disease relation (9,388 edges), with 3,170 oncology diseases flagged by an ontology-keyword filter.

Node type	Count	dim	Node type	Count	dim
biological_process	28,642	384	disease	17,080	384
gene / protein	27,610	384	effect_phenotype	15,311	384
anatomy	14,033	384	molecular_function	11,169	384
drug	7,957	384	pathway	2,516	384
cellular_component	4,176	384	exposure	818	384

Table 1: PrimeKG node inventory. Every node carries a 384-d SentenceTransformer embedding of its name, shared identically by the GNN and the tabular baselines so that any performance gap measures graph structure, not features.

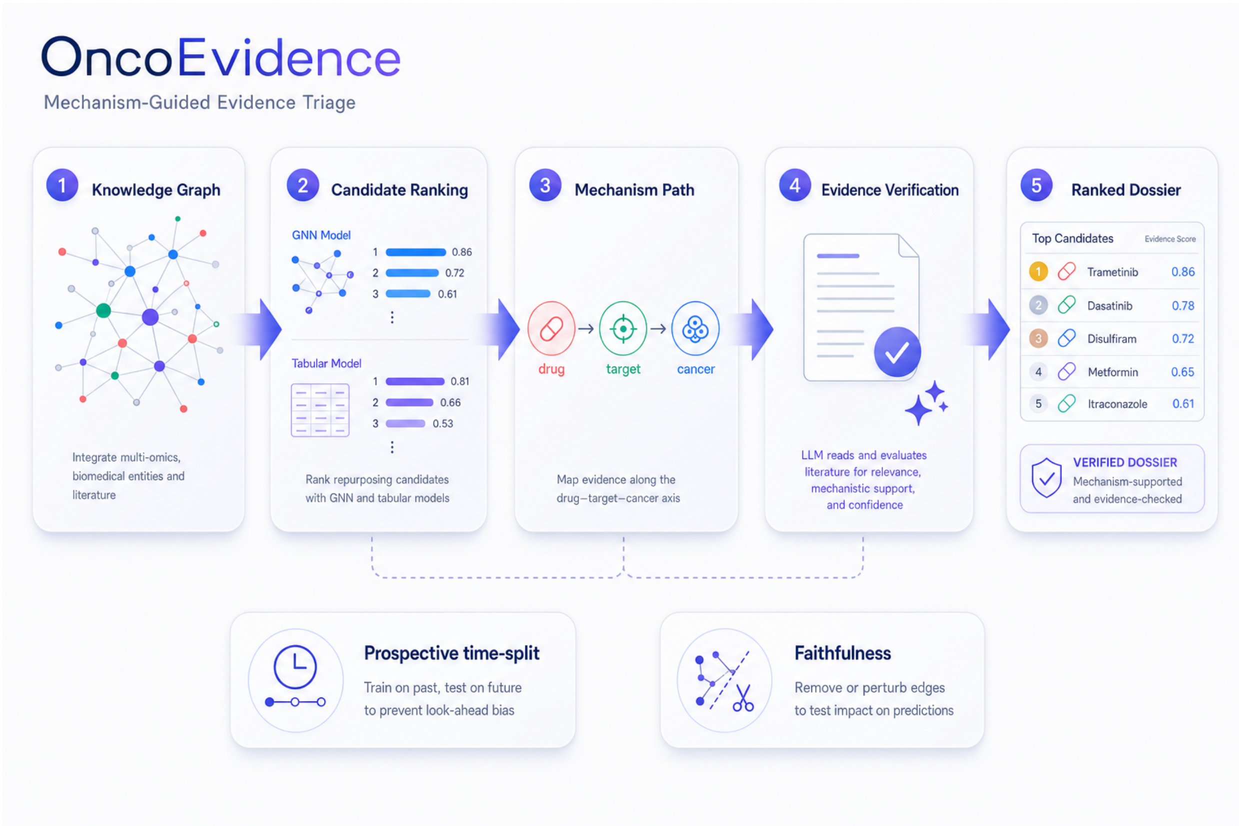

Figure 1: The OncoEvidence pipeline. We rank drug–disease candidates over PrimeKG, extract a multi-hop mechanism path (drug $\rightarrow$ target $\rightarrow$ cancer gene $\rightarrow$ cancer), verify it against Europe PMC abstracts with an LLM, and emit an evidence-backed ranked dossier. Two audits stress-test the system: a prospective time-split (train on past, test on future) and a counterfactual faithfulness test (remove mechanistic edges and re-evaluate).

3 Methods

3.1 Candidate ranking and the controls

Repurposing is framed as link prediction on the drug $\rightarrow$ indication $\rightarrow$ disease relation. A heterogeneous GraphSAGE [6] encoder builds an embedding for every node by message passing over its typed neighbours, and an MLP decoder scores a pair:

$$\mathbf{h}_v^{(l)} = \sigma\Big(W^{(l)}\big[\,\mathbf{h}_v^{(l-1)} \,\|\, \mathrm{AGG}_{u\in\mathcal{N}_r(v)}\,\mathbf{h}_u^{(l-1)}\big]\Big), \qquad s(d,c) = \mathrm{MLP}\big(\big[\,\mathbf{z}_d \,\|\, \mathbf{z}_c\,\big]\big). \tag{1}$$

Three controls receive the *same* features: a tuned XGBoost on the concatenated endpoint features (the primary tabular ranker), a FeatureMLP (the same projection and decoder but no message passing), and a DistMult knowledge-graph embedding $s(d,c) = \mathbf{z}_d^\top \mathrm{diag}(\mathbf{r})\,\mathbf{z}_c$ (a pure memorizer with no representation for unseen nodes). All models are trained with binary cross-entropy on positives and sampled negatives.

3.2 Mechanism-path extraction

For a candidate we search PrimeKG for short paths through mechanism relations (drug-target, protein-protein, protein-pathway, protein-disease), scoring each by specificity and penalizing promiscuous hubs:

$$\mathrm{score}(p) \;=\; \beta_{\text{type}} \;+\; \underbrace{\frac{1}{\log_2(\deg(\text{bridge})+2)}}_{\text{bridge specificity}} \cdot \underbrace{\frac{1}{\log_2(\delta_{\text{drug}}+2)}}_{\text{IDF carrier penalty}}, \tag{2}$$

where δ_{drug} is the number of drugs that bind the bridge protein (an inverse-frequency penalty that suppresses generic carriers such as albumin) and β_{type} orders direct-target (3) over interaction (2) over shared-pathway (1) templates. A typical recovered chain:

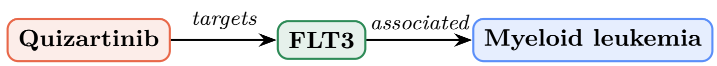

Figure 2: A direct-target mechanism path. The extractor recovers textbook MOAs such as Quizartinib–FLT3, Tamibarotene–RARA, Topotecan–TOP1, and Trifluridine–TYMS for true indications; random pairs usually yield no path.

3.3 The mechanism head and the joint objective

A mechanism head scores a (drug, gene, disease) triple on the *same* encoder embeddings, $s_{\text{mech}}(d, g, c) = \mathrm{MLP}\big([\mathbf{z}_d \| \mathbf{z}_g \| \mathbf{z}_c]\big)$. We train it jointly with the link objective using an InfoNCE term that ranks the curated DrugMechDB bridge gene g^+ above K degree-matched decoys g_k^-:

$$\mathcal{L} = \mathcal{L}_{\text{link}} + \lambda\,\mathcal{L}_{\text{mech}}, \qquad \mathcal{L}_{\text{mech}} = -\log \frac{\exp s_{\text{mech}}(d, g^+, c)}{\exp s_{\text{mech}}(d, g^+, c) + \sum_{k=1}^{K} \exp s_{\text{mech}}(d, g_k^-, c)}. \tag{3}$$

3.4 The two audits: prospective and faithfulness

Prospective (temporal) split. PrimeKG has no timestamps, so we assign each true oncology pair (d, c) an approximate first-evidence year $\tau(d, c)$ = earliest Europe PMC co-mention. For a cutoff T, the PAST set $\{(d, c) : \tau \le T\}$ seeds the message-passing graph and supervision, while the FUTURE set $\{(d, c) : \tau > T\}$ is held out; its target edges are removed, so the model must rank future indications above negatives using only pre-T structure.

Counterfactual faithfulness. For a held-out pair whose bridge gene g is a real drug-target edge e_{dg}, we recompute g's mechanism score after deleting e_{dg} and compare against deleting a random other target edge e_{dr} of the same drug:

$$\Delta_{\text{MOA}} = s_{\text{full}} - s_{\setminus e_{dg}}, \quad \Delta_{\text{rand}} = s_{\text{full}} - s_{\setminus e_{dr}}, \quad \text{contrast} = \overline{\Delta_{\text{MOA}}} - \overline{\Delta_{\text{rand}}}. \tag{4}$$

A faithful model satisfies $\Delta_{\text{MOA}} \gg \Delta_{\text{rand}}$.

3.5 Metrics

We report AUROC and AUPRC for separation/ranking, recall@k for retrieval, and the precision of "supported" verdicts,

$$\text{precision} = \frac{\#\{\text{“supported” pairs whose path gene is in the DrugMechDB MOA set}\}}{\#\{\text{“supported” pairs DrugMechDB covers}\}}. \tag{5}$$

4 Results

4.1 Finding 1: the link task does not need the graph

With a tuned XGBoost on shared features, the GNN does not win in any regime (Table 2, Fig. 3). Paired tests confirm it: in the cold-disease regime XGBoost beats the GNN by 0.081 AUROC (the GNN-favoring one-sided test gives $p = 0.98$), while the GNN crushes the DistMult memorizer (+0.43 AUROC, $p = 2.4 \times 10^{-4}$). For link ranking with rich text features, a strong tabular model suffices.

Test AUROC (5 seeds)	GNN	XGBoost	MLP	KGE
Transductive	0.977	**0.988**	0.973	0.907
Inductive (cold-disease, oncology)	0.882	**0.963**	0.930	0.452
Inductive (cold-drug)	0.956	**0.958**	0.931	0.482

Table 2: Corrected, leakage-safe benchmark (mean over 5 seeds). The DistMult memorizer (KGE) collapses on unseen nodes in both inductive regimes, confirming the task needs content features, not memorized identity.

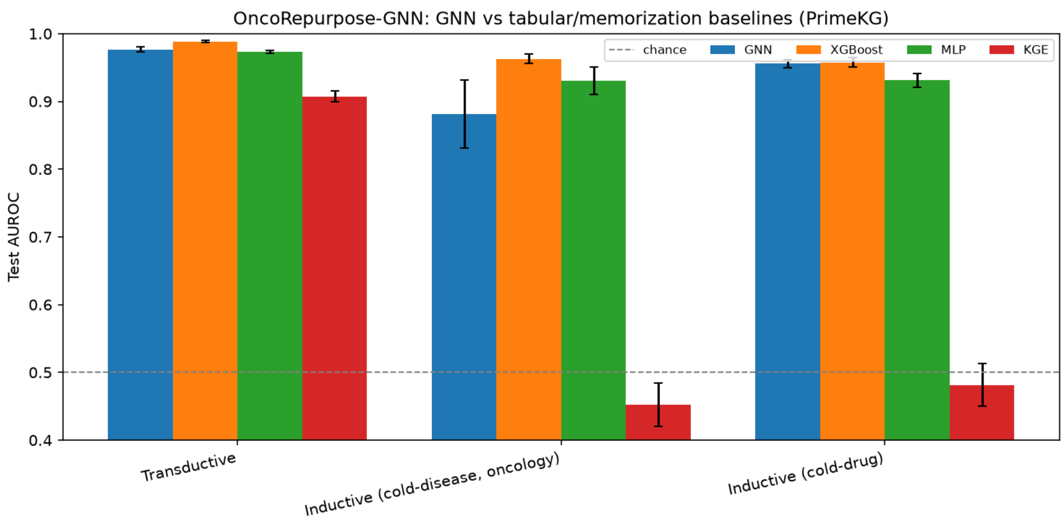

Figure 3: Benchmark AUROC by regime and model (5 seeds, error bars = std). XGBoost (or a tie) leads everywhere; KGE collapses on the inductive splits.

4.2 Ablations: where the GNN's ranking signal comes from

Three ablations locate the GNN's signal (Table 3, Fig. 4). Removing all auxiliary topology (*empty*) drops cold-disease AUROC from 0.899 to 0.777, but *shuffling* edges barely hurts (0.879), so the GNN leans on node features and coarse connectivity more than on specific graph structure. Dropping individual relation families (drug-protein, disease-protein, pathway) each costs little. And under non-semantic *structural* features the GNN still does not beat XGBoost (0.838 vs 0.861), so there is no hidden cold-start ranking win to claim.

Topology (cold-disease)	GNN AUROC
Intact graph	0.899
Shuffled edges	0.879
Empty (features only)	0.777

Relation drops: drug-protein 0.934, disease-protein 0.920, pathway 0.933.

Features	GNN	XGB	MLP
Text (384-d)	0.889	**0.960**	0.939
Structural (23-d)	0.838	**0.861**	0.830
Random (64-d)	0.837	0.490	0.843

Table 3: Left: topology and relation ablations (3 seeds, cold-disease). Right: feature ablation (2 seeds). XGBoost's collapse to chance under random features confirms it uses real signal; the GNN never overtakes it.

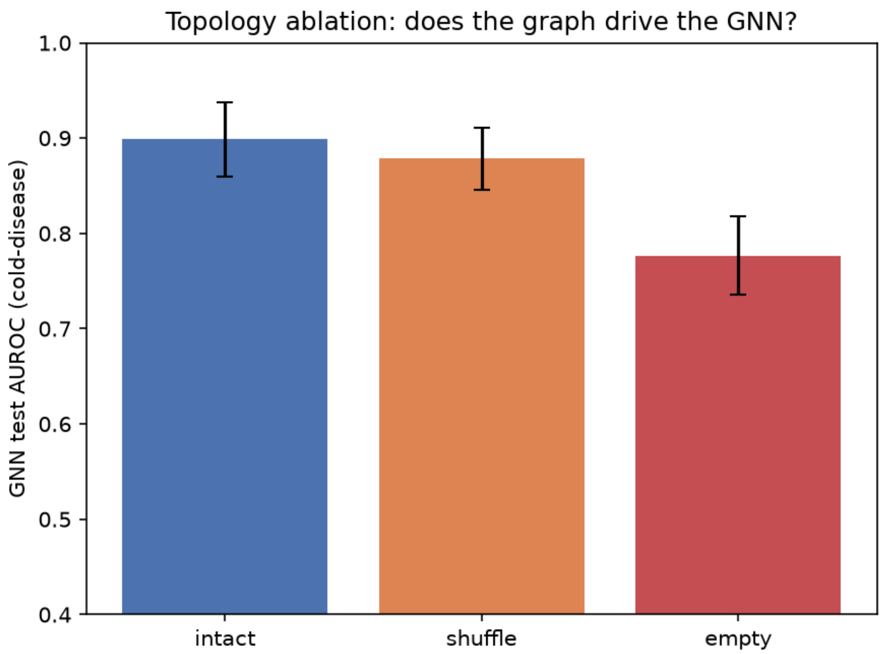

Figure 4: Topology ablation. Erasing structure (*empty*) costs the most; shuffling edges costs little, indicating the GNN's ranking relies mostly on node features.

4.3 Finding 2–3: the graph carries real, discriminative mechanism

A tabular model cannot produce a traceable mechanism; the graph can (Fig. 2). Over 400 true indications versus 400 random pairs, the mechanism signal separates the two groups at **AUROC 0.879** (Table 4, Fig. 5), and the extracted bridge genes agree with curated DrugMechDB for 211 of 263 covered pairs (**0.802**; e.g. Methotrexate→{ATIC, DHFR, TYMS}, Topotecan→{TOP1}, Decitabine→{DNMT1, DNMT3A}).

Metric	True	Random
Mean mechanism score	**1.95**	0.16
Direct-target rate	**34.0%**	0.25%
Any mechanistic path	**81.3%**	8.8%

Separation AUROC
0.879

Table 4: Mechanism paths separate true indications from random pairs. A true indication is roughly 136× more likely to have a direct drug-target to cancer-gene link.

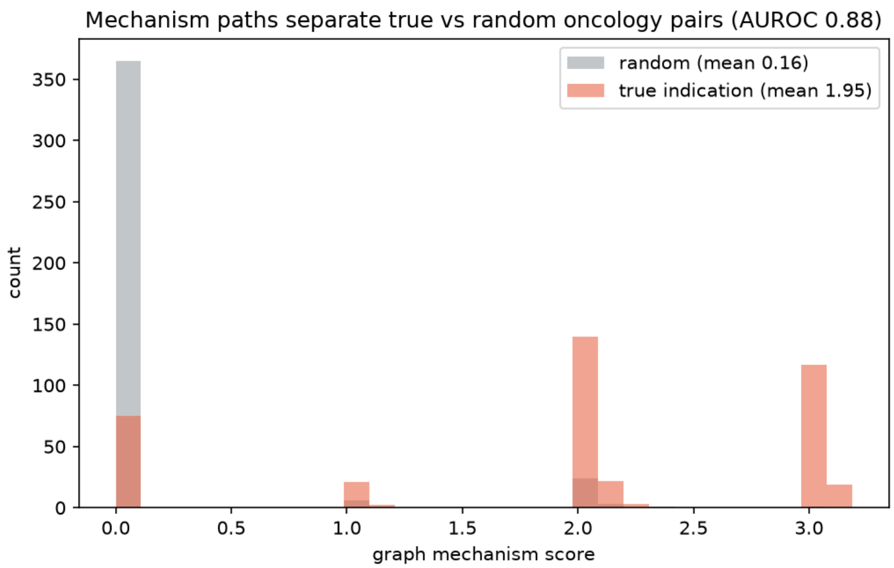

Figure 5: Distribution of the graph mechanism score for true indications versus random drug-cancer pairs. Random pairs pile up at zero; true indications spread to high, specific scores (AUROC 0.879).

Robustness to hard negatives. Separation degrades gracefully as negatives get harder and only the deliberately adversarial *shared-target* case nears chance (Table 5). An ablation that removes *direct*-target paths still separates (0.887), so the signal is not merely the most obvious drug-target link.

Negatives vs true	random	oncology-drug	degree-matched	shared-target
Separation AUROC	0.887	0.870	0.742	0.609

Table 5: Hard-negative stress test (400 vs 400, seed 0). Ablations: full 0.887, template-only 0.884, no-direct-target 0.887, so indirect mechanisms also separate.

The verifier is a conservative, precise reviewer. Over 50 true versus 50 random pairs the language model marks far fewer "supported" than a keyword baseline, but those calls are more trustworthy (Table 6): of its supported calls DrugMechDB covers, 6/7 are confirmed versus 13/22 for keyword matching.

	Language model	Keyword baseline
True pairs graded "supported"	11 / 50	34 / 50
Random pairs graded "supported"	1 / 50	2 / 50
"Supported" confirmed by DrugMechDB	**0.857** (6/7)	0.591 (13/22)

Table 6: Verifier precision (OpenRouter `gpt-4o-mini`). Small denominators (Wilson 95% CI roughly 0.49–0.97 for 6/7), so this is a pilot signal. Each verdict carries a verbatim quote, e.g. Topotecan / TOP1: "a topoisomerase I (TOP1) inhibitor".

4.4 Finding 4: the graph's one genuine ranking-free edge

We trained the GNN jointly (Eq. 3) and asked it to name the curated bridge gene for held-out cold-disease pairs (3 seeds, $n \approx 186$–260). Unblinded, a trivial "the drug's own targets" baseline dominates because the bridge gene is usually the drug's direct target. But with that direct edge hidden, the trivial and degree baselines collapse to about zero while the joint GNN still recovers the gene from indirect structure (Table 7). A link-only GNN scored through the identical head recovers nothing, so the signal is attributable to the auxiliary objective, not the architecture.

Bridge-gene recovery	Unblinded R@10	Blinded R@10	Blinded MRR
Joint GNN (link + mechanism)	0.31	**0.25**	**0.21**
Trivial target-lookup	**0.80**	0.00	0.00
Link-only GNN (same head, no aux loss)	0.00	0.00	0.00
Degree prior	0.02	0.02	0.02

Table 7: Mechanism recovery on held-out pairs. When the direct edge is hidden, only the joint GNN recovers the gene, a small (∼0.25 R@10) but genuine graph-only signal. Caveat: recovery of *known* mechanism via indirect paths, not de novo discovery, on a modest sample.

4.5 Finding 5 (new): the model is predictive over time

To test whether the model could have flagged an indication *before* it was established, we hold out indications by time (cutoff $T = 2005$; 350 sampled pairs, 341 years resolved, 240 past / 101 future; 3 seeds). Trained only on pre-2005 structure, the GNN ranks the held-out future indications at **AUROC 0.930** and **AUPRC 0.707** and beats the structure-blind control on every metric and seed (Table 8, Fig. 6). Unlike the plain link task in Finding 1, the graph adds a small but consistent edge here, widest on the harder metrics.

T=2005	AUROC	AUPRC	R@50	R@100
GNN (graph)	**0.930**	**0.707**	**0.393**	**0.703**
MLP (blind)	0.882	0.592	0.337	0.587
Graph gain	+0.047	+0.115	+0.056	+0.116

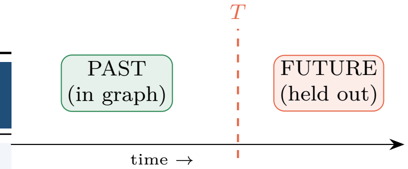

Table 8: Prospective time-split (mean over 3 seeds). PAST pairs seed the graph; FUTURE pairs are predicted from pre-T structure only. Caveat: the year proxy is the earliest literature co-mention, not a regulatory date.

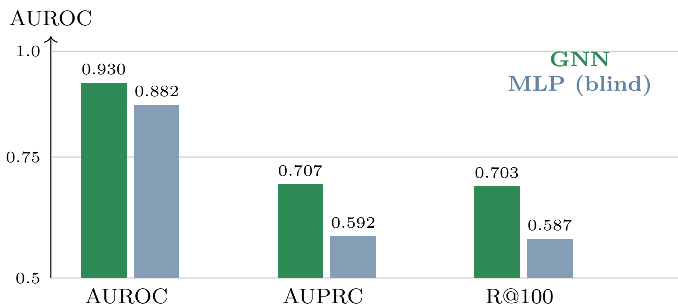

Figure 6: Prospective performance: the graph model (green) beats the structure-blind control (blue) on every metric, with the gap widening on the harder AUPRC and recall@100.

From a metric to named cases. The AUROC hides what the model actually surfaces, so we read off, for each held-out *future* indication, where its drug landed when we score every drug against that cancer using only pre-T structure. Of 101 future indications, the GNN placed **11 in the top 50** of ∼7,956 candidate drugs — a **17.3**× enrichment over a random screen — against **1** for the structure-blind control (1.6×), so the graph yields 11× as many top-50 prospective hits and puts 13.9% of future indications in the top ∼1% versus 1.0% for the control. The hits are recognizable post-cutoff oncology approvals the model had no indication edge for (Table 9). The graph's edge is concentrated where a shortlist matters — the very top of the ranking; across the bulk of pairs the structure-blind control is competitive, which we report rather than hide.

Drug	Cancer (future indication)	1st lit. yr	GNN rank	MLP rank
Trabectedin	ovarian carcinoma (several subtypes)	2006–10	**2–5**	3038
Methoxsalen	cutaneous T-cell lymphoma	2006	**10**	2154
Romidepsin	cutaneous T-cell lymphoma	2008–10	**63**	1726
Pralatrexate	cutaneous T-cell lymphoma	2009	**321**	2235
Nilotinib	CML, BCR-ABL1 positive	2006	**342**	1098
Plerixafor	plasma cell myeloma	2007	**428**	1797

Table 9: Prospective named cases (cutoff $T = 2005$, single seed). Each indication is first co-mentioned in the literature after T, yet a model trained on pre-T structure alone ranks the drug near the top of ∼7,956 candidates for that cancer, far above the structure-blind control. Ranks are model- and cutoff-dependent; the year is the earliest Europe PMC co-mention, not a regulatory date.

4.6 Finding 6 (new): the mechanism reasoning is faithful

Can we trust the mechanism the graph names? For each held-out pair whose bridge gene is a real drug-target edge we apply Eq. 4: recompute the gene's score with the full graph, with that pair's curated mechanism edge deleted, and with a random other target edge of the same drug deleted (316 instances, 2 seeds, no retraining). Removing the curated edge degrades the bridge gene far more than removing a random one (Table 10, Fig. 7): a paired Wilcoxon test gives $p \approx 3.6 \times 10^{-34}$, the model is faithful in 83% of instances, and the two populations separate at AUROC 0.835.

Counterfactual edge removal	Score drop	Rank drop
Remove curated mechanism edge	**0.124**	∼**211**
Remove random target edge (same drug)	−0.001	∼2
Contrast (faithfulness)	**0.124**	—

Table 10: Faithfulness by counterfactual edge removal (pooled, 316 instances; logit space). Faithful in 83.3% of cases; paired Wilcoxon $p \approx 3.6 \times 10^{-34}$; separation AUROC 0.835. Measured on the well-posed population (pairs whose bridge gene is an actual drug-target edge).

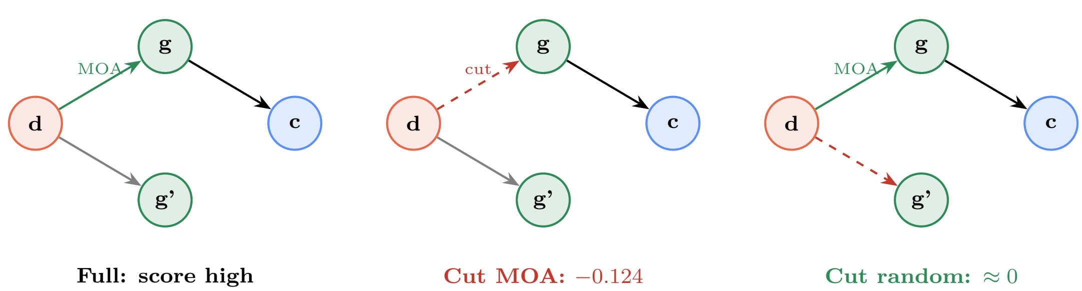

Figure 7: Counterfactual faithfulness. Cutting the curated MOA edge $d \to g$ collapses the bridge gene's score; cutting a random other target edge $d \to g'$ barely changes it.

4.7 An honest, fully characterized negative: orthogonal trial validation

We also asked whether top *novel* predictions (not already PrimeKG indications) appear as registered oncology trials more than matched controls, and we chased the confounds all the way down. Top raw-score predictions were not enriched over random (6.7% versus 7.4%, Fisher $p = 0.69$). Raw score does track trial existence above chance (AUROC 0.676 over 810 pairs, $p \approx 1.5 \times 10^{-5}$), but a popularity confound is visible immediately: one broadly-indicated drug (folic acid) accounts for 78% of the top set's hits.

Conditioning on the drug (comparing only cancers for the same drug, which cancels drug popularity) the model looks strong — within-drug stratified AUROC 0.821 ($p = 5 \times 10^{-4}$ by within-stratum permutation; 49 drugs, 2,649 cancer pairs), and for 31/49 drugs the single highest-scored cancer has a real trial (base rate 0.39, $p = 5 \times 10^{-4}$). But this still confounds *cancer* popularity (e.g. cervical cancer scores ≈ 0.998 for almost every drug and is widely trialed). Removing *both* marginals with a two-way fixed-effect model — $\mathrm{score}(d, c) = \mu + \alpha_d + \beta_c + \varepsilon_{dc}$ — and testing whether the interaction residual ε_{dc} predicts a trial gives **AUROC** 0.475 **(structure-aware permutation** $p = 0.85$**): no signal**. So the apparent within-drug effect was cancer popularity, not pairwise specificity. The honest reading: raw deployment scores carry drug- and cancer-level popularity but not the drug×cancer specificity this registry could corroborate. This is a clean, well-understood negative rather than a muddy one, and it stands in deliberate contrast to the prospective result (Finding 5), where specific predictive signal does appear.

5 Discussion

The contribution is a disciplined account of where a knowledge graph helps. On plain link ranking it does not beat a tuned tabular model, and we say so. Its value is the traceable mechanism a tabular model cannot produce, and this report strengthens that case along the two axes that matter for a real triage tool. Finding 5 shows the mechanism-aware graph model is *predictive*: trained only on pre-2005 structure, it ranks post-2005 indications at 0.930 AUROC and beats the structure-blind control, so the signal is not merely retrospective bookkeeping. Finding 6 shows the mechanism reasoning is *faithful*: deleting the curated edge, and only that edge, collapses the prediction, so the explanation reflects the biology the model actually used.

Limitations. The blinded recovery and faithfulness tests are recovery of known mechanism, not de novo discovery; some samples are modest; the time axis is a literature proxy; and the orthogonal trial check is a negative, though now a fully characterized one (the signal is drug/cancer popularity, not pairwise specificity). None of the predictions are medical advice.

6 Reproducibility

Open tools (PyTorch Geometric, scikit-learn, XGBoost) and public data (PrimeKG, Europe PMC, DrugMechDB, ClinicalTrials.gov). Every reported number has a script and a JSON record in `results/`: `oncorepurpose.json` (Findings 1 + ablations), `feature_ablation.json`, `mechanism_eval.json` (Finding 3), `hard_negatives_eval.json`, `verify_llm_eval.json`, `mechanism_recovery_eval.json` (Finding 4), `temporal_split_eval.json` and `prospective_case_studies.json` (Finding 5, including the named cases), `edge_faithfulness_eval.json` (Finding 6), and `clinicaltrials_validation.json` with `clinicaltrials_deconfounded.json` (the characterized negative). A six-part Colab notebook track reproduces the pipeline end to end, with Part 4 fully self-contained. All predictions are hypothesis-generating research, not medical advice.

References

[1] Huang K, Chandak P, et al. *A foundation model for clinician-centered drug repurposing (TxGNN)*. Nature Medicine, 2024.

[2] Chandak P, Huang K, Zitnik M. *Building a knowledge graph to enable precision medicine (PrimeKG)*. Scientific Data, 2023.

[3] *KGML-xDTD: a knowledge graph-based framework for drug treatment prediction and mechanism description*. 2023.

[4] Zitnik M, Agrawal M, Leskovec J. *Modeling polypharmacy side effects with graph convolutional networks (Decagon)*. Bioinformatics, 2018.

[5] Mayers M, et al. *DrugMechDB: a curated database of drug mechanism paths*. 2022.

[6] Hamilton W, Ying R, Leskovec J. *Inductive representation learning on large graphs (GraphSAGE)*. NeurIPS, 2017.